Project - High Level Design

On

Non Profit Quiz /Tutor Bot

Course Name:

Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Academic Year: 2026

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1. Introduction

1.1 Scope of the document

The purpose of this document is to describe the design, architecture, and functionality of the Generative AI-based Non-Profit Quiz /Tutor Bot. This document outlines the overall system structure, application workflow, data flow, and technical components used to build the solution.

The system is designed to assist users in learning about the non-profit sector through an interactive quiz-based educational platform powered by Generative Artificial Intelligence and Large Language Models (LLMs). The bot evaluates user responses, provides contextual explanations for incorrect answers, and helps reinforce learning through intelligent feedback.

This document covers the system design, process flow, component architecture, API integrations, and design considerations required to implement the application. It serves as a technical reference for developers, researchers, and stakeholders involved in the development and maintenance of the system.

1.2 Intended Audience

This document is intended for individuals involved in the design, development, deployment, and evaluation of the Non-Profit Quiz Tutor Bot system. The primary audience includes:

- Software Developers and Engineers who will implement and maintain the system architecture.
- AI/ML Engineers responsible for integrating Large Language Models and vector databases for intelligent responses.
- Researchers and Students studying Generative AI applications in education and non-profit domains.
- Non-Profit Organizations interested in adopting AI-based learning tools for training and awareness.

The document provides both technical and conceptual information to help these audiences understand how the system operates and how its components interact.

1.3 System Overview

The Non-Profit Quiz Tutor Bot is an AI-driven educational platform designed to help users improve their knowledge of non-profit sector concepts through an interactive learning experience.

The system presents users with quiz questions derived from donor communications, fundraising strategies, and non-profit operational concepts. After the user submits an answer, the system evaluates the response using a Large Language Model (LLM). Instead of only indicating whether the answer is correct or incorrect, the AI generates detailed explanations and contextual feedback to help users understand the correct concept.

The system also uses a Vector Database to store knowledge related to non-profit topics, allowing the AI model to retrieve relevant information and provide accurate explanations. The chatbot interface enables users to interact with the system naturally, making the learning experience more engaging.

Key technologies used in the system include:

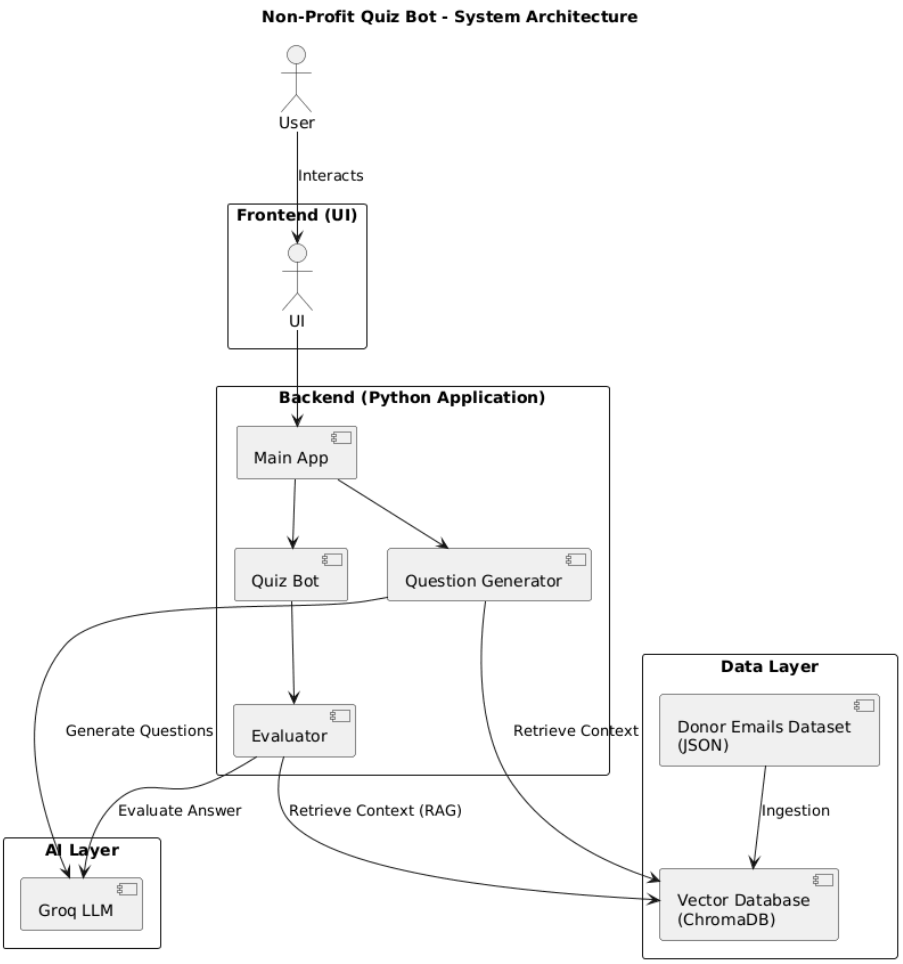
- **Python** for backend development
- **Large Language Models (LLMs)** for intelligent evaluation and explanation
- **Vector Databases (FAISS/ChromaDB)** for semantic search
- **HTML, CSS and JavaScript** for building the chatbot interface
- **API integrations** for connecting the application with AI models

2. System Design

2.1 Application Design

The Non-Profit Quiz/Tutor Bot is designed as a modular AI-driven system that processes donor email data to generate and evaluate learning assessments. The application begins by loading donor emails from a JSON dataset, which are then split into smaller chunks and converted into embeddings for storage in a vector database (ChromaDB). This enables efficient semantic retrieval of relevant information. The question generation module uses a Large Language Model (LLM) via the Groq API to create context-based quiz questions from the stored data. During the quiz, the user interacts with the system through the quiz bot module, which displays questions and collects responses. The evaluator module then analyzes

the user's answers using a combination of direct matching (for objective questions) and LLM-based evaluation (for descriptive answers). It also retrieves relevant email context using a Retrieval-Augmented Generation (RAG) approach to generate accurate and meaningful explanations. The system is controlled by a main controller that manages data ingestion, quiz flow, and evaluation, thereby creating a personalized learning loop that helps users improve their understanding of nonprofit communication practices.



The Non-Profit Quiz Tutor Bot follows a layered application architecture consisting of four main layers: Frontend (User Interface), Backend (Python Application), AI Layer, and Data Layer. Each layer performs a specific function and communicates with other components to deliver an interactive AI-based learning experience. The architecture ensures modularity, scalability, and efficient data processing.

2.1.1. User Interface (Frontend Layer)

The Frontend Layer acts as the interaction point between the user and the system. Users access the application through a chatbot-style interface where they can answer quiz questions and receive explanations.

The interface is designed to be simple and interactive so that users can easily engage with the learning system. It sends user responses to the backend for processing and displays the feedback generated by the AI system. Technologies such as Streamlit or web-based UI frameworks can be used to build this layer.

The main responsibilities of the frontend include:

- Accepting user responses to quiz questions
- Displaying quiz questions generated by the system
- Showing evaluation results and AI-generated explanations
- Managing user interaction with the chatbot interface

2.1.2. Backend Layer (Python Application)

The Backend Layer is the core processing unit of the application. It manages the logic required for generating quiz questions, evaluating answers, and retrieving contextual knowledge from the database.

The backend consists of several modules:

- **Main Application Module:** The main application module controls the overall workflow of the system. It manages communication between the frontend, quiz engine, evaluation system, AI model, and database.

- **Quiz Bot Module:** The Quiz Bot module manages quiz interactions. It sends quiz questions to the user, collects answers, and forwards them for evaluation.

- **Question Generator:** The evaluator analyzes the user's answers and determines whether they are correct or incorrect. It sends the response to the AI model for deeper analysis and explanation generation.

2.1.3. AI Layer

The AI Layer is responsible for intelligent reasoning and explanation generation. This layer integrates a Large Language Model (LLM) **Groq API - LLaMA 3.1** to perform natural language understanding and generate contextual responses.

The AI model performs the following functions:

- Evaluates the correctness of user answers
- Generates explanations for incorrect answers
- Provides contextual feedback to enhance learning
- Uses retrieved knowledge to ensure accurate responses

This layer uses a Retrieval-Augmented Generation (RAG) approach, where the AI model retrieves relevant information from the vector database before generating responses. This improves accuracy and reduces hallucinations.

2.1.4. Data Layer

The Data Layer manages the storage and retrieval of knowledge used by the system. It contains structured and processed information related to non-profit sector topics.

The main components of this layer include:

- **Donor Emails Dataset (JSON):** The dataset contains donor communication examples and non-profit related content stored in JSON format. This dataset is used to train and support the quiz generation process.
- **Vector Database (ChromaDB):** The dataset is converted into embeddings and stored in a vector database. This allows the system to perform semantic search and retrieve relevant context when evaluating answers or generating questions.

The vector database supports the **Retrieval-Augmented Generation (RAG)** mechanism, enabling the AI model to access accurate contextual information before generating explanations.

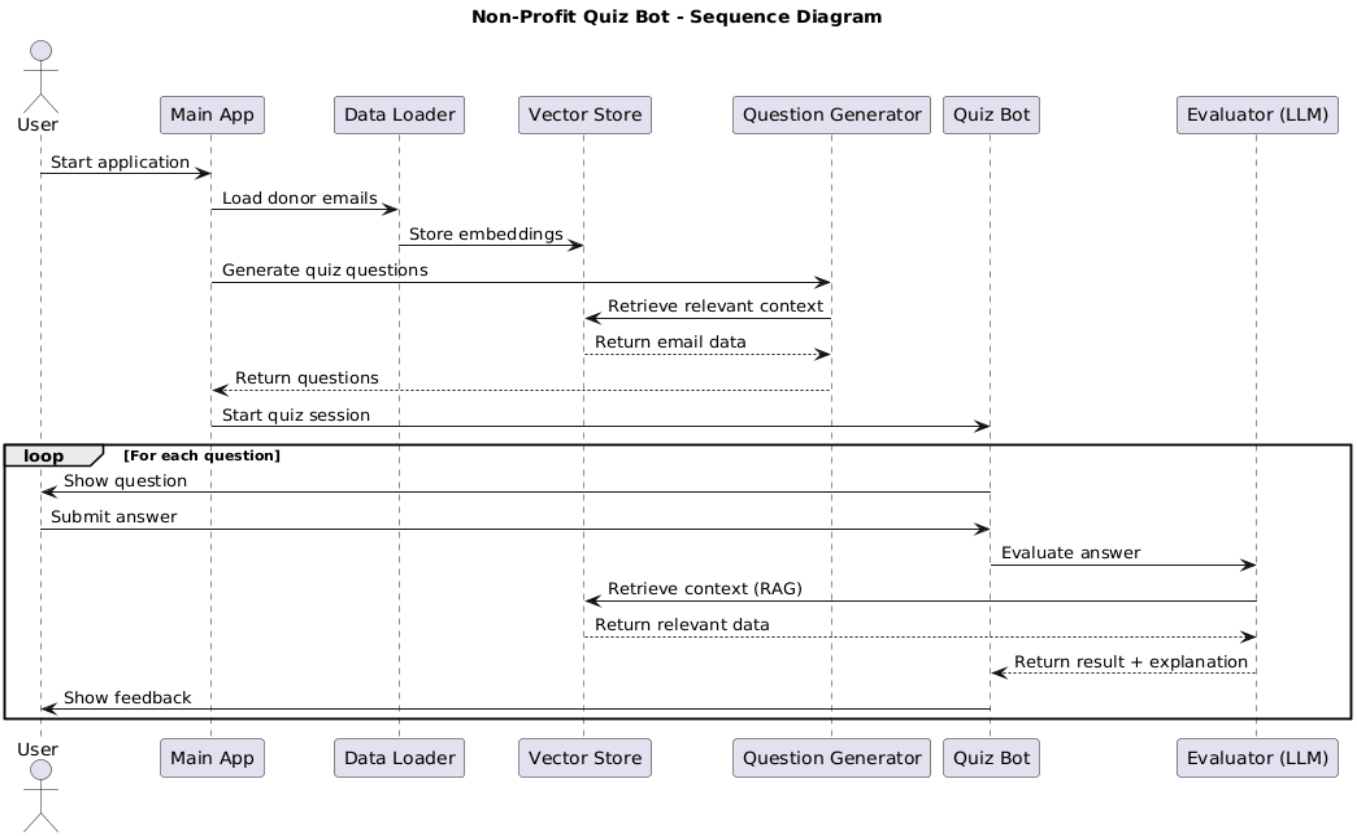
2.1.5. System Workflow

The interaction between different components follows a structured workflow:

1. The user interacts with the frontend interface and submits an answer.
2. The frontend sends the response to the backend main application.

3. The Quiz Bot module manages the quiz session and forwards the answer to the Evaluator module.
4. The evaluator retrieves relevant context from the Vector Database (ChromaDB) using semantic search.
5. The retrieved information is sent to the Groq LLM in the AI layer.
6. The AI model evaluates the answer and generates an explanation.
7. The result is returned to the backend and displayed to the user interface.

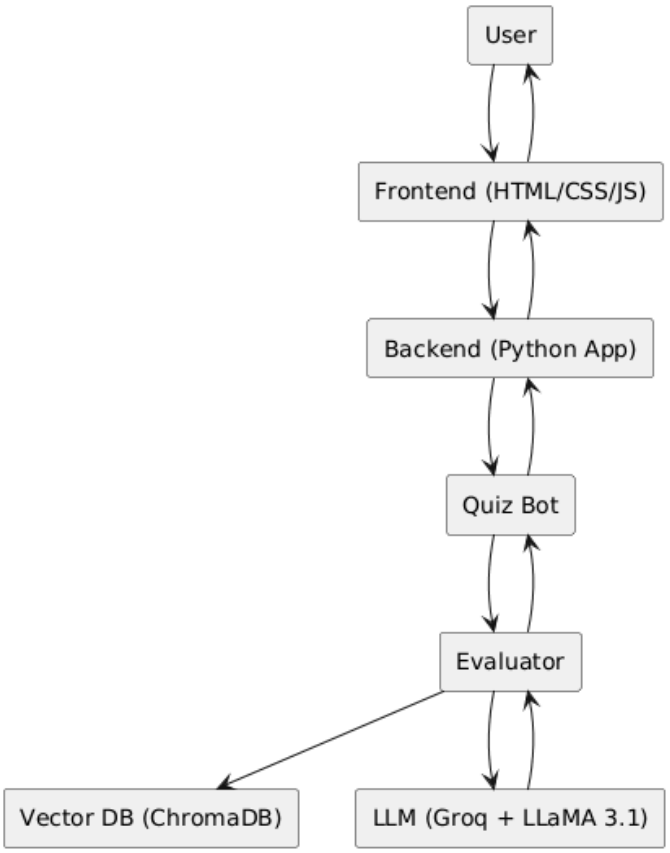
2.2 Process Flow



Sequence Diagram

- User starts the quiz from the interface
- System generates questions using the Question Generator
- Question Generator retrieves relevant data from the Vector Database
- Questions are sent to the user via the Quiz Bot
- User submits an answer
- Quiz Bot forwards the answer to the Evaluator
- Evaluator checks correctness (logic + LLM)
- Relevant context is retrieved from the Vector Database (RAG)
- Context and answer are sent to the LLM (Groq + LLaMA)
- LLM generates explanation/feedback
- Result is returned to Quiz Bot
- Final feedback is displayed to the user

2.3 Information Flow



2.4 Components Design

The system consists of several components that work together to deliver the final functionality

.

- **Main Application (main.py):** Acts as the entry point of the system and controls the overall workflow. It manages user modes such as quiz, tutor, and ingestion, and coordinates interaction between all components.

- **Quiz Bot:** Handles user interaction by displaying questions, collecting responses, and showing results. It connects the user interface with backend logic and ensures smooth quiz flow.

- **Question Generator:** Responsible for generating quiz questions using the LLM. It retrieves relevant data from the vector database and creates structured questions of different types like MCQ, True/False, and short answer.

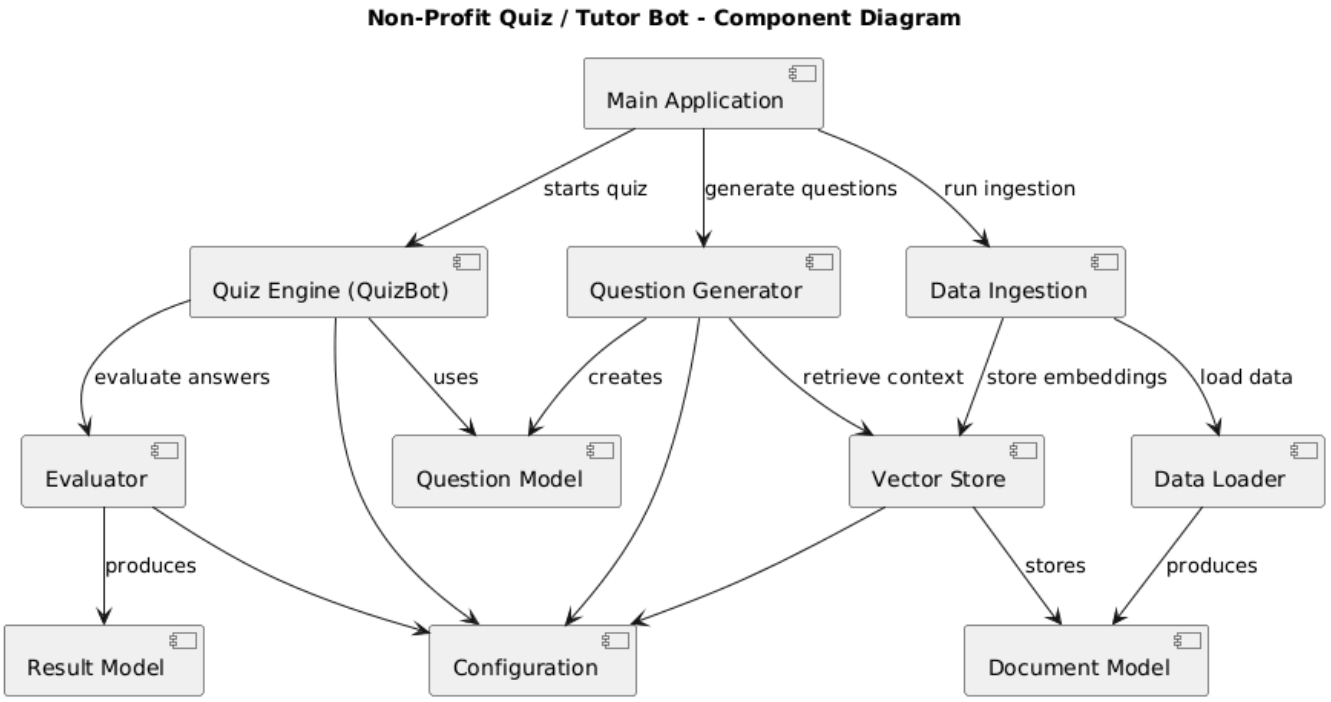
- **Evaluator:** Evaluates user responses using both direct comparison (for objective questions) and LLM-based analysis (for descriptive answers). It also generates contextual explanations using the RAG approach.

- **Data Loader:** Loads donor email data from JSON files and converts it into structured document format for further processing.

- **Vector Store (ChromaDB):** Stores embeddings of donor email content and enables semantic search. It retrieves relevant information for question generation and evaluation, forming the core of the RAG system.

- **Ingestion Module:** Processes raw data by performing chunking and embedding generation, and stores it into the vector database. This step prepares the data for efficient retrieval.

- **LLM (Groq API + LLaMA 3.1):** Performs key AI tasks such as generating questions, evaluating answers, and providing explanations. It works with retrieved context to ensure accurate and meaningful responses.



Component Diagram

2.5 Key Design Considerations

- **Scalability**
  The system architecture allows easy expansion to support more users and larger knowledge bases.
- **Accuracy of AI Responses**
  The use of a vector database ensures that the AI retrieves reliable and relevant knowledge.
- **User Experience**
  The chatbot interface is designed to be simple, interactive, and user-friendly.
- **Modularity**
  Different components are separated so that updates can be implemented without affecting the entire system.
- **Security and Privacy**
  User data and responses are handled securely to ensure privacy and data protection.
- **Performance Efficiency**
  Efficient retrieval mechanisms and optimized AI queries ensure fast response times.

2.6 API Catalogue

API Function Name	Module	Input	Output	Purpose
load_donor_emails()	Data Loader	JSON file path	List of Documents	Loads donor email dataset
chunk_documents()	Data Loader	List of Documents	Chunked Documents	Splits emails into parts
build_vector_store()	Vector Store	Document chunks	Vector DB created	Stores embeddings in ChromaDB
similarity_search()	Vector Store	Query, k	Relevant Documents	Retrieves related email context
generate_quiz()	Question Generator	Number of questions	List of Questions	Generates quiz using LLM
evaluate_answer()	Evaluator	Question, User Answer	Result (correct/incorrect + explanation)	Evaluates answers using LLM
run_ingestion()	Ingestion	None	Vector DB ready	Executes full data pipeline
run_quiz_session()	Quiz Bot	List of Questions	Score + Feedback	Runs quiz interaction

3. Data Design

3.1. Data Model

The system uses a semi-structured data model based on JSON files containing donor emails. Each email includes fields such as *id, subject, sender, date, and body*.

During processing, these emails are converted into document objects and further divided into smaller chunks. These chunks are then transformed into vector embeddings and stored in a vector database (ChromaDB) for semantic retrieval.

3.2. Data Access Mechanism

The system uses a Retrieval-Augmented Generation (RAG) approach for data access.

When a query is made:

- The system performs semantic search on the vector database
- Retrieves the most relevant email chunks
- Passes them to the LLM (via Groq API)

This ensures fast and context-aware access to relevant information instead of scanning the entire dataset.

3.3. Data Retention Policies

The system stores:

- Donor email dataset (JSON file)
- Vector embeddings in ChromaDB

Data is stored locally and retained only for the duration of the project. No sensitive user data is permanently stored.

For scalability, retention policies can be extended to include:

- Periodic cleanup of unused data
- Secure storage and access control

3.4. Data Migration

Data migration involves moving raw email data into the vector database.

1. Load donor emails from JSON
2. Convert into document format
3. Split into chunks
4. Generate embeddings
5. Store in ChromaDB

This process is handled by the ingestion pipeline and needs to be run only once unless new data is added.

4. Session and State Management

4.1. Session Management

The application uses a temporary session model where each user interaction session begins when the user accesses the web interface and initiates a quiz or tutor mode. Session continuity is maintained on the client side using browser memory (such as JavaScript variables or session storage), while the backend processes each request independently. No persistent session storage or authentication mechanism is implemented, making the system lightweight and suitable for single-user interaction scenarios.

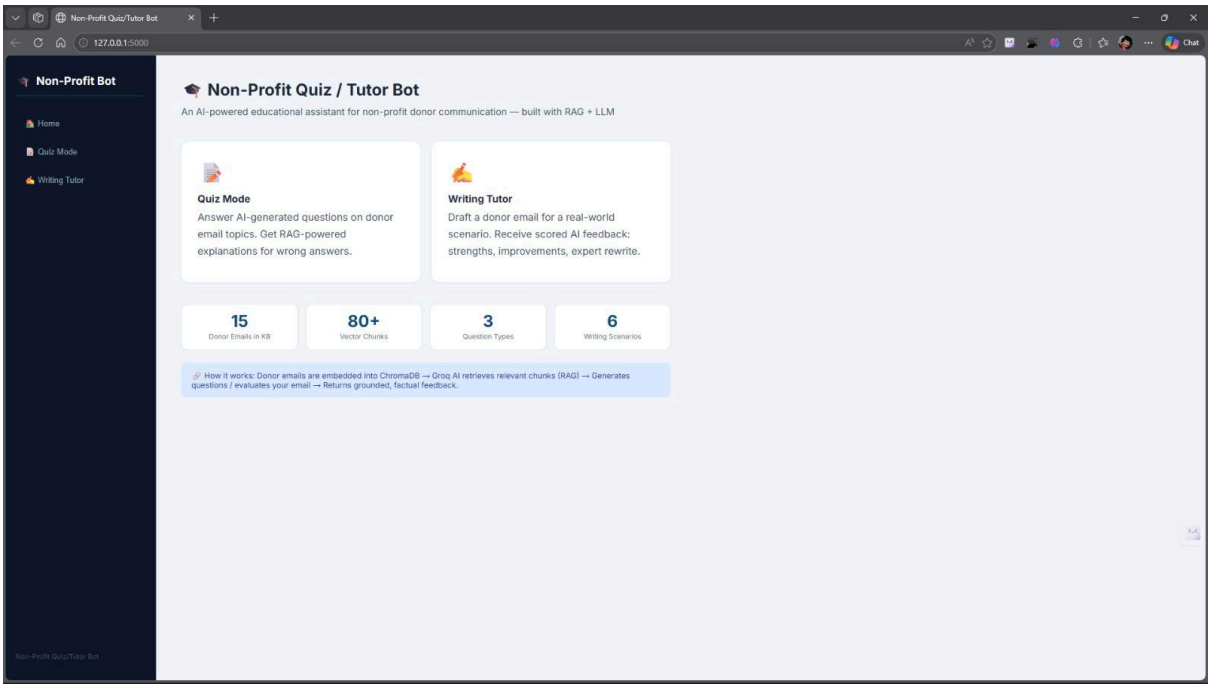
4.2. State Management

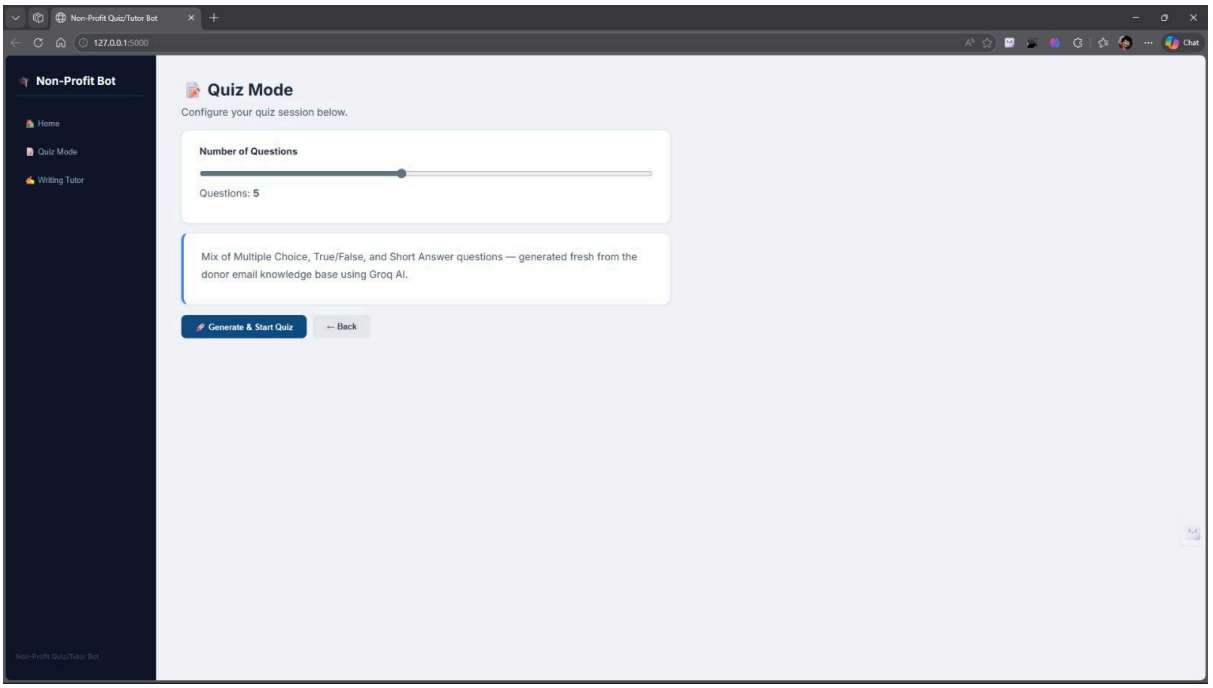
State is managed across both frontend and backend components. On the frontend, JavaScript maintains the current state, including the current question, user responses, and navigation between questions. On the backend, state is handled in-memory during request processing, including question generation, evaluation results, and contextual explanations. The Quiz Bot logic coordinates state updates, while the evaluator processes each response and returns feedback dynamically. No long-term state persistence is used.

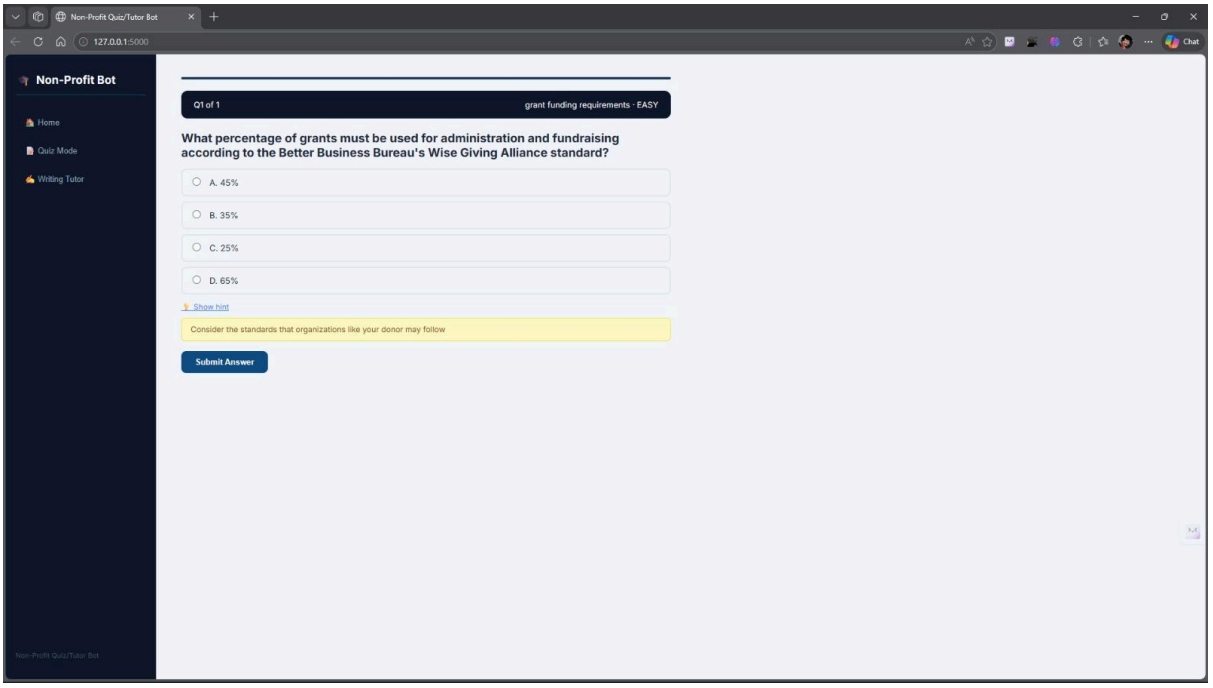
4.3. Frontend–Backend Interaction

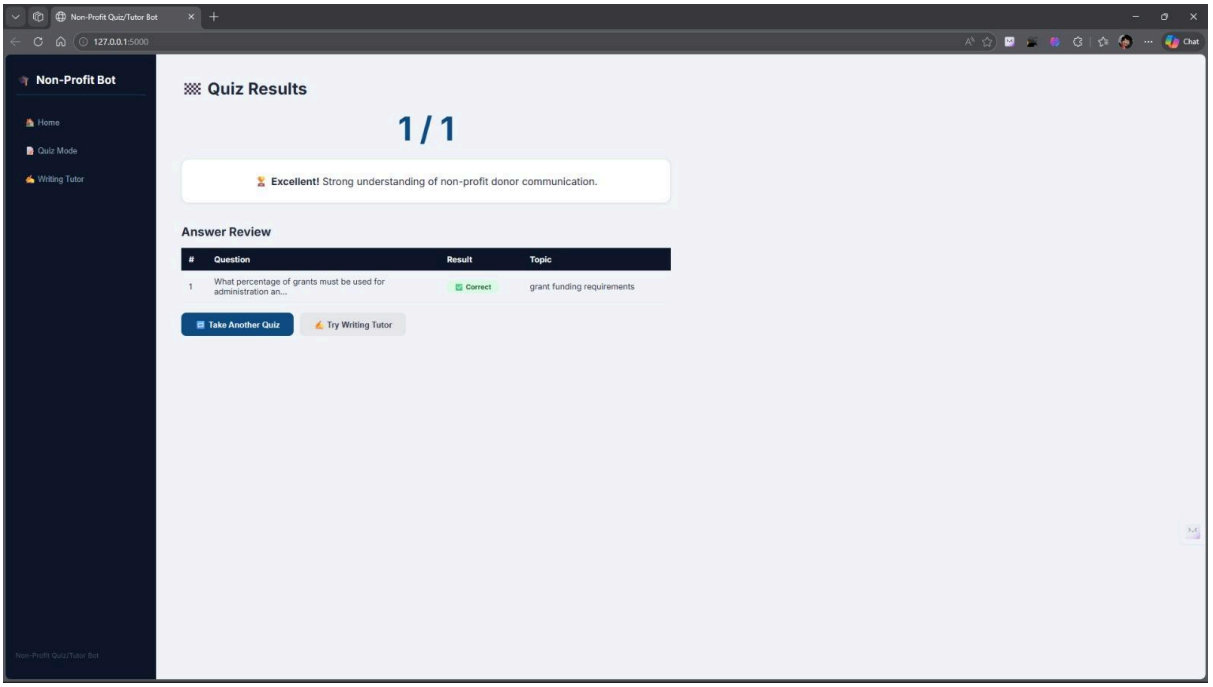
The frontend communicates with the backend through API calls, sending user responses and receiving evaluation results. Each request is stateless from the backend perspective, meaning the server does not retain session data between requests. The frontend is responsible for maintaining continuity of the user experience by storing and updating session-related information.

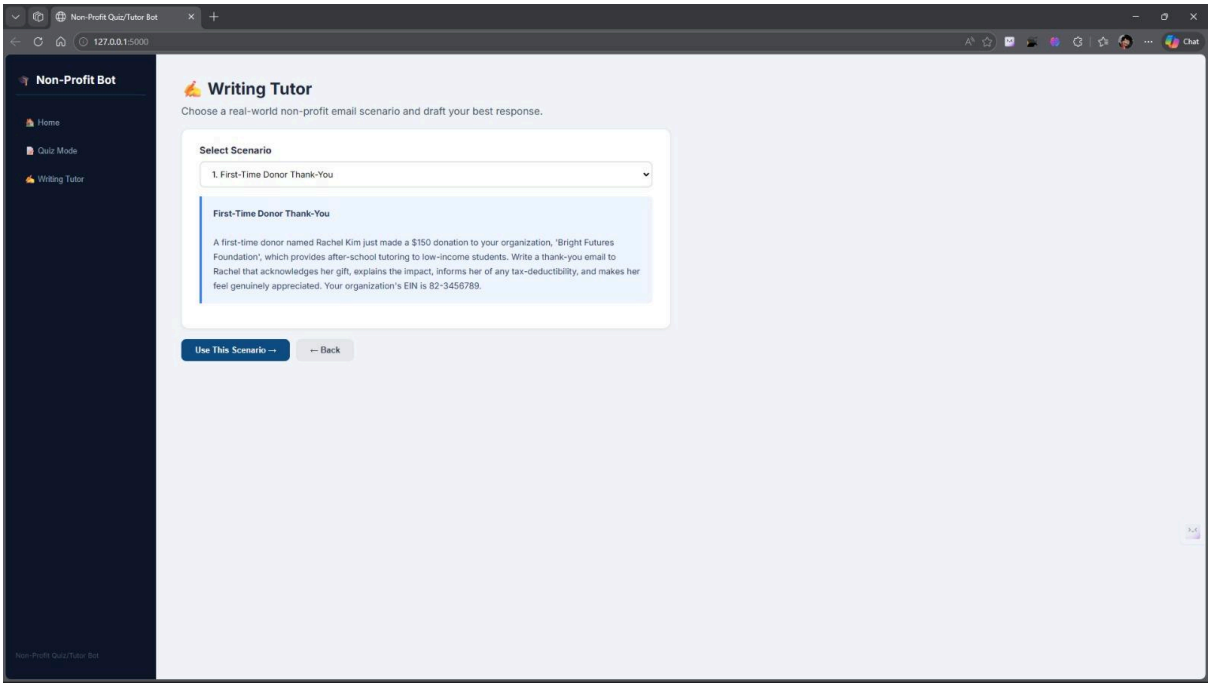
5. Interfaces

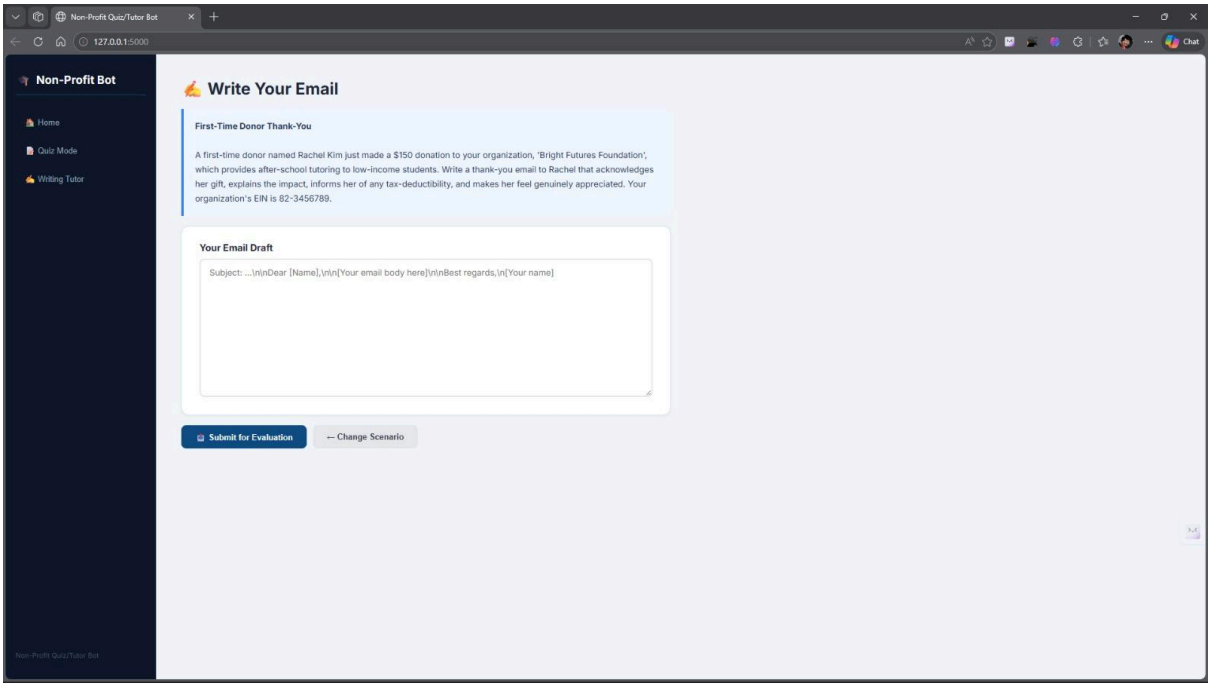


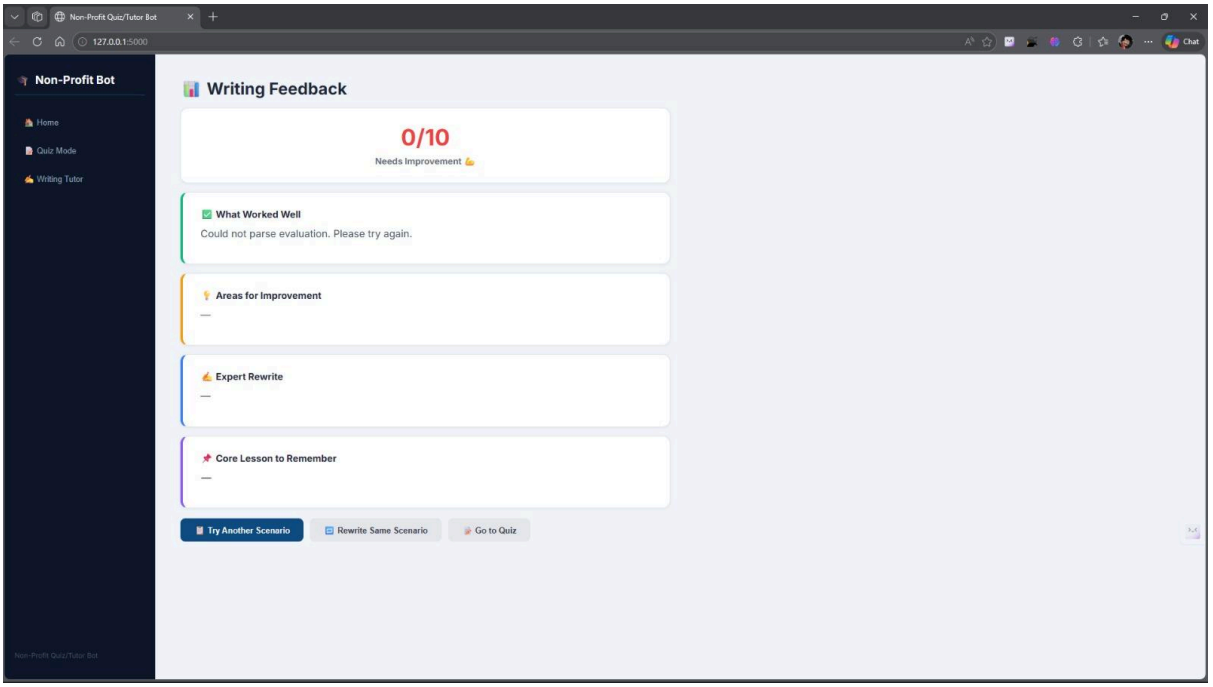


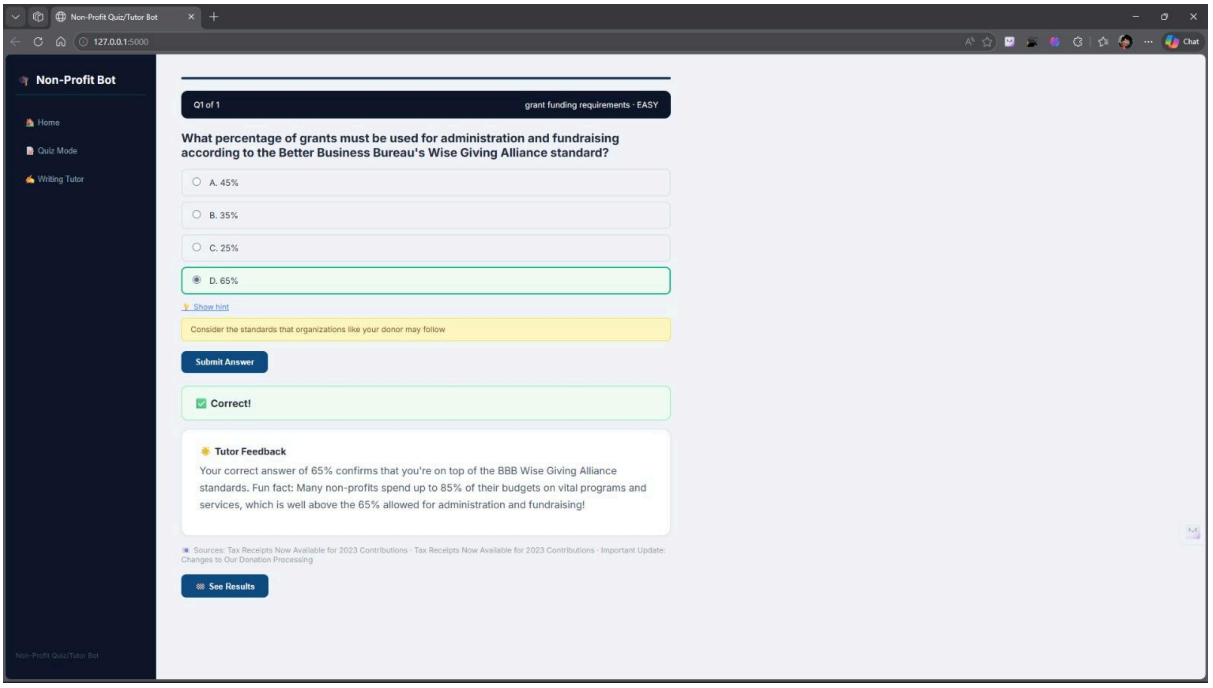

Non-Profit Quiz/Tutor Bot
127.0.0.1:5000
Non-Profit Bot
Home
Quiz Mode
Writing Tutor
Q1 of 1
grant funding requirements · EASY
What percentage of grants must be used for administration and fundraising according to the Better Business Bureau's Wise Giving Alliance standard?
A. 45%
B. 35%
C. 25%
D. 65%
Show hint
Consider the standards that organizations like your donor may follow
Submit Answer


Non-Profit Quiz/Tutor Bot
127.0.0.1:5000
Non-Profit Bot
Home
Quiz Mode
Writing Tutor
Quiz Results
1 / 1
Excellent! Strong understanding of non-profit donor communication.
Answer Review
Question Result Topic
1 What percentage of grants must be used for administration an... Correct grant funding requirements
Take Another Quiz
Try Writing Tutor


Non-Profit Quiz/Tutor Bot
127.0.0.1:5000
Non-Profit Bot
Home
Quiz Mode
Writing Tutor
Writing Tutor
Choose a real-world non-profit email scenario and draft your best response.
Select Scenario
1. First-Time Donor Thank-You
First-Time Donor Thank-You
A first-time donor named Rachel Kim just made a $150 donation to your organization, 'Bright Futures Foundation', which provides after-school tutoring to low-income students. Write a thank-you email to Rachel that acknowledges her gift, explains the impact, informs her of any tax-deductibility, and makes her feel genuinely appreciated. Your organization's EIN is 82-3456789.
Use This Scenario →
← Back


Non-Profit Quiz/Tutor Bot
127.0.0.1:5000
Non-Profit Bot
Home
Quiz Mode
Writing Tutor
Write Your Email
First-Time Donor Thank-You
A first-time donor named Rachel Kim just made a $150 donation to your organization, 'Bright Futures Foundation', which provides after-school tutoring to low-income students. Write a thank-you email to Rachel that acknowledges her gift, explains the impact, informs her of any tax-deductibility, and makes her feel genuinely appreciated. Your organization's EIN is 82-3456789.
Your Email Draft
Subject: ...\n\nDear [Name],\n\n[Your email body here]\n\nBest regards,\n[Your name]
Submit for Evaluation
← Change Scenario


Non-Profit Quiz/Tutor Bot
127.0.0.1:5000
Non-Profit Bot
Home
Quiz Mode
Writing Tutor
Writing Feedback
0/10
Needs Improvement
What Worked Well
Could not parse evaluation. Please try again.
Areas for Improvement
Expert Rewrite
Core Lesson to Remember
Try Another Scenario
Rewrite Same Scenario
Go to Quiz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Writing Tutor
Q1 of 1
grant funding requirements · EASY
What percentage of grants must be used for administration and fundraising according to the Better Business Bureau's Wise Giving Alliance standard?
A. 45%
B. 35%
C. 25%
D. 65%
Show hint
Consider the standards that organizations like your donor may follow
Submit Answer
Correct!
Tutor Feedback
Your correct answer of 65% confirms that you're on top of the BBB Wise Giving Alliance standards. Fun fact: Many non-profits spend up to 85% of their budgets on vital programs and services, which is well above the 65% allowed for administration and fundraising!
Sources: Tax Receipts Now Available for 2023 Contributions · Tax Receipts Now Available for 2023 Contributions · Important Update: Changes to Our Donation Processing
See Results

6. Caching

The system implements in-memory caching for the vector database (ChromaDB) to improve performance. The vector store is loaded once during application startup and stored in a global variable, allowing it to be reused across multiple requests without repeated disk access. This reduces latency during semantic search operations and enhances overall system efficiency. However, caching is limited to the vector store, and LLM responses are generated dynamically for each request.

7. Non-Functional Requirements

7.1. Security Aspects

- The system shall store API keys securely using environment variables (.env file) and avoid hardcoding sensitive information.
- The system shall not store any personally identifiable user data, ensuring basic user privacy.
- The backend shall follow a stateless architecture to minimize session-related vulnerabilities.
- The system shall restrict direct access to internal data sources such as the vector database.
- Advanced security features like authentication, authorization, and encryption are not implemented and can be considered as future enhancements.

7.2. Performance Aspects

- The system shall provide fast response times for quiz generation and evaluation.
- The vector database shall be cached in memory to reduce retrieval latency.
- The system shall use Retrieval-Augmented Generation (RAG) to fetch only relevant data, improving efficiency.
- The system shall minimize API calls by using batch question generation.
- The performance may depend on external LLM API response time, and further optimization can be achieved using caching and asynchronous processing.

8. References

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